

ETC3460 / ETC5346

Assignment 2

Due: Wednesday 27 May 2026 (Week 12), 11:55 PM

Assignment 2 must be submitted electronically in PDF format. It will be marked out of 90, and this mark will be converted to a mark out of 20 marks for your final grade. Your file needs to be uploaded by only one member of each group. However, all group members must click the "Submit Assignment" button on Moodle and accept the University's submission statement. This is essential, so please make sure to do this. Late submission of your work will result in a loss of marks.

Each group member will also be required to complete an anonymous peer evaluation survey. The survey will be done via Feedback Fruits in Moodle. You will be asked to rate your group members' participation and effort. These surveys may also be used to adjust your assignment marks (for example, if one group member is deemed to only have contributed half as much as everyone else, they will only get half of the assignment mark). Failure to complete the survey will incur a penalty of 5 marks. Further details will be released via Moodle in due course.

Your answer to part A should be a maximum of 10 pages, with paragraphs clearly separated. Please use a font size of at least 11 pt throughout the assignment. Important graphs and equations (in equation form, not screenshots of R output) must be included within the report. You do not need to provide R code or an Appendix.

Submission instructions:

Please submit your PDF file via Moodle. You must submit your answers in PDF format, rendered from a Quarto (or similar) document. You may create the output in another format (such as a Word or HTML document) as an intermediate step, but you must convert it to PDF for submission in a neat and presentable format.

Introduction

This assignment is to be completed with your corresponding group members. Students should attempt Part A Questions 1 and 2 themselves before consulting with their group members. **Part A Question 3 and Part B encourage collaboration among group members from the outset.** Full credit requires answering all questions.

Part A

Question 1 [25 marks = 3+(2+4)+4+5+3+4]

All tests should be conducted at the 5% significance level. Clearly label each step when testing a hypothesis. Download daily data from Yahoo Finance for Alphabet Inc. over the period from 5 February 2016 to 26 March 2026. Compute the daily log returns using adjusted closing prices and **express the log returns in percentages**.

```
getSymbols(Symbols = c("GOOG"), from="2016-02-05", to="2026-03-27",  
auto.assign=TRUE, warnings=FALSE)
```

*Note: Although the required sample period ends on 26 March 2026, the **to** date in `getSymbols()` is set to 27 March 2026 to ensure that the observation for 26 March 2026 is included in the Yahoo Finance download.*

- a. Report the autocorrelations through lag 10 for the squared log returns. What do these autocorrelations tell you about the time-series properties of GOOG returns?
- b. Using the log-return series, estimate both an AR(1) and an AR(2) specification for the returns. Compare the two fitted models and choose the specification that you consider more appropriate. In your answer, briefly explain the evidence supporting your choice.
- c. For your chosen model in part b, conduct an ARCH test with $p = 5$ lags. What conclusion do you draw?
- d. Add an ARCH(3) component to your chosen mean equation from part b and re-estimate the model (use a normal error distribution). Comment on the coefficient estimates in the variance equation.
- e. Run a GARCH(1,1) model and comment on its coefficient estimates.
- f. Determine whether the ARCH(3) or GARCH(1,1) model provides the better fit to the data, using the same mean equation as in part b. Clearly state the statistics that you use and present all required steps.

Hint: You can collect your numerical results in a table and graphical results in one image to allow space for commentary in your report. The equations and testing steps can be described once.

Question 2 [20 marks = 4+3+3+3+2+5]

You may submit a handwritten solution for Question 2 only. It must be clearly legible, not reduced in size or shrunk in the image, and included in your single PDF submission. Show all working. Assume y_t is stationary. Consider the AR(1) model

$$y_t = 0.9 + 0.7y_{t-1} + \varepsilon_t$$

where ε_t is a martingale difference sequence (MDS). Let F_{t-1} denote the information set available up to time $t - 1$.

(a) Using the method of successive substitution, show that

$$y_t = 0.9 \sum_{j=0}^{k-1} 0.7^j + 0.7^k y_{t-k} + \sum_{j=0}^{k-1} 0.7^j \varepsilon_{t-j} \quad (1)$$

for $k \geq 1$.

(b) As $k \rightarrow \infty$, show that the model in equation (1) can be written as

$$y_t = \frac{0.9}{1 - 0.7} + \sum_{j=0}^{\infty} 0.7^j \varepsilon_{t-j}. \quad (2)$$

(c) Using equation (2), what is $E(y_t)$? Provide a numerical answer.

(d) Derive the 1-step-ahead forecast $E(y_{t+1}|F_t)$, 2-step-ahead forecast $E(y_{t+2}|F_t)$ and 3-step-ahead forecast $E(y_{t+3}|F_t)$.

(e) Based on your answers to part (d), deduce that the h -step-ahead forecast is given by

$$E(y_{t+h}|F_t) = 0.9(1 + 0.7 + \dots + 0.7^{h-1}) + 0.7^h y_t$$

(f) For any $a \neq 1$, the geometric sum formula is

$$1 + a + \dots + a^{h-1} = \frac{1 - a^h}{1 - a}.$$

Show that the h -step-ahead forecast is given by

$$E(y_{t+h}|F_t) = \mu + 0.7^h (y_t - \mu)$$

where $\mu = E(y_t)$. Hence, find

$$\lim_{h \rightarrow \infty} E(y_{t+h}|F_t).$$

Question 3 [10 marks = 7+3]

In the Week 6 workshop, we examined the following US-listed Vanguard ETFs arranged in a 3×3 style box:

	Growth	Blend	Value
Small	VBK	VB	VBR
Mid	VOT	VO	VOE
Large	VUG	VV	VTV

Using monthly adjusted closing prices from Yahoo Finance over the period December 2015 to December 2025, we found that the evidence from these ETFs over this 10-year sample does not support the classic Fama–French size and value patterns. In particular, the mean monthly log-return style box showed that VUG had the highest average monthly return. A student therefore suggests: “Perhaps VUG is the ETF we should *all* buy.” An industry practitioner replies: “Not so fast. A higher average return does not necessarily make an investment the best choice. In finance, **there is no free lunch**. We need to ask whether higher returns are associated with higher risk, and it may also depend on an investor’s risk preferences; e.g. a risk-averse investor may prefer an ETF with lower risk, even if its average return is lower”. Use this discussion as motivation for answering the following questions.

(a) Using the same nine ETFs and the same sample period, construct a second 3×3 style box in which each cell reports one numerical risk measure for the corresponding ETF. Each ETF should therefore have a numerical risk value. Choose one appropriate measure of risk based on the concepts studied in this unit. You must: (i) clearly state the risk measure you have chosen; (ii) define it carefully; (iii) explain why it is appropriate in this context; and (iv) calculate it correctly. Present your results in the same 3×3 format (style box) used for the mean monthly log-return table in the workshop.

Note: *This is an open-ended, subjective question with no single correct answer. Marks will be awarded for the quality of your justification.*

(b) Using your risk table from part (a), together with the mean monthly log-return 3×3 style box from the workshop, discuss whether the observed return differences, if any, across the ETFs appear to be associated with differences in risk over this sample period. In particular, comment on whether the evidence suggests a risk-return trade-off. For example, where ETFs have higher mean monthly log-returns, do they also tend to have higher measured risk?

Note: *There is no single correct conclusion. Marks will be awarded for the quality of your interpretation and your use of evidence from comparing the risk-table in part (a) with the return-table from the workshop. The purpose of this question is to give you practical experience in analysing ETFs using ideas from quant (factor) investing.*

Part B [35 marks]

Your manager needs to present your work to the Investment Committee of your firm. Your group has been tasked with preparing a slide deck of up to 13 slides. You must defend your portfolio's performance and the econometric methodology used to forecast its returns. The Investment Committee values statistical rigour and expects a semi-technical business presentation style. Your manager will review your slides before the presentation. You are not required to present.

Slide 1: Title Slide

Slide 2: Executive Summary

Up to two slides: In Assignment 1 Part B, you used data up to 23 March 2026 to construct a portfolio of your chosen stocks. Using the portfolio weights obtained on 23 March 2026 (the weights must remain unchanged), analyse the performance of this portfolio over the period from 24 March 2026 to 8 May 2026. Provide a time-series plot of the daily portfolio returns over this period, report summary statistics for your portfolio, and evaluate the overall performance of your portfolio.

Up to seven slides: Using data from 24 March 2026 to 1 May 2026 alone would not be sufficient for the following analysis. You are required to use all available historical data up to 1 May 2026. Using concepts learnt in class, identify the time series properties of the daily portfolio returns. You should specifically address the following points:

- a. Estimate an appropriate mean equation.
- b. Do they have ARCH effects? If yes, you are expected to model this component of the data.
- c. Back up your discussion for parts (a) and (b) with detailed methodology learned throughout the semester. You should also discuss at least one other competing model which you examined but, in the end, chose not to use.
- d. Using the final model selected in part (c), generate forecasts of your portfolio returns during the last week of the investment exercise (4 May to 8 May 2026) and discuss your results. Your forecasts should be generated using only information available up to 1 May 2026.

Up to two slides: Discuss the limitations of your analysis and outline how the approach could be improved or conducted differently.